Benjamin Öztürk

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EDUCATION

ESPCI PARIS PSL | Paris, France *Sept 2024 – Present*
Master Student
Optics, Quantum Physics, Electromagnetic Waves, Acoustic Waves, Fluid Mechanics, Material Mechanics, Crystallized Materials, Electronics, Statistical Mechanics, Deep Learning.

Preparatory Class PCSI-PC* | Nantes, France *Sept 2022 – July 2024*
Lycée Clémenceau
Intensive 2 year scientific program to prepare for nationwide engineering school exams.

British International Section | Nantes, France *Sept 2019 – Jul 2022*
Lycée Blanche de Castille

- High School Diploma with High Honors. Specialties : Mathematics, Physics.
- Participated in the 2022 *Concours Général de Mathématiques* (French National High School Maths Competition)

EXPERIENCE

Intern Scientist | Procter & Gamble, Smart Products *Jul 2026 – Dec 2026*
German Innovation Center, Kronberg, Germany
My work focuses on in-vitro thermal modeling of hair growth reduction by light. Combine theoretical modeling of heat transfer equations in biological systems with experimental laser calibration and high-speed thermal imaging.

- Currently developing a setup having human skin-like thermal properties. Measurement of the temperature profiles with high speed IR Camera and comparison with FEM simulated data will ensue.
- Elaborating an experimental setup with layers of dermis and fat phantoms mimicking the optical properties of the skin. Temperature profiles under laser pulses have been measured with thermocouples.

Intern | ESPCI SIMM Laboratory-Sciences and Engineering of Soft Matter *Sept 2024*
Paris, France

- Collaborated with a PhD Researcher with his research on the effects of cryopreservation on cells.
- Synthesized lipid vesicles and microfluidic chips for mimicking living cells and studying osmotic phenomena during freezing.
- Used confocal microscopy to quantify deformation and mechanical stress on the vesicle during freezing.

Intern | Paris Saclay C2N Laboratory- Center for Nanosciences and Nanotechnologies *Oct 2021*
Saclay, France

- Assisted a Senior Researcher investigating quantum dots and contributed to the research of two PhD Candidates developing biomedical microfluidic applications.
- Learned to visualize quantum dots using Scanning Electron Microscopy (SEM).
- Fabricated silicon wafers masks for microfluidic chips with photolithography technique in clean room.
- Synthesized PDMS microfluidic chips for the fabrication of an artificial lung prototype.

PROJECTS

French Chemistry Tournament - 4th place *Sept 2025 – March 2026*
Participated as Team ESPCI and conducted a challenge aiming at chemically identifying coriander in a random dish and suppressing its taste. My work was rated as "impressive" (grade $\geqslant 25/30$).

- Identified the volatile aldehydes responsible for the coriander aroma by SPME-GC-MS.
- Developed a protocol for the encapsulation of volatile aldehydes by cyclodextrines. The targeted aldehydes were complexed in aqueous solution by ß-cyclodextrines and the complexation rate was computed via H-NMR and SPME-GC-MS.

The use of chalk in rock climbing *Jan 2023 – June 2024*

Investigated the adhesion properties of Magnesium Carbonate (chalk) in rock climbing to address the ambiguity in scientific literature regarding its physical efficacy versus placebo effect.

- Elaborated an experimental setup for measuring the static coefficient of friction between skin and climbing holds. Conducted the experiments with and without chalk.
- Determined and compared the efficiency of both liquid and solid chalk in adherence to the holds.

SKILLS & TECHNICAL TOOLS

Languages: French (native), English (C2 Cambridge CAE), German (B1), Turkish (basics)

Computer: Python (Pytorch, NumPy, SciPy), Matlab, ImageJ.

Lab skills:

- Optics and Laser Systems: Laser alignment and calibration, Wavefront manipulation (SLM), Infrared imaging, Fabry-Perot interferometer.
- Microscopy and Spectroscopy: Confocal, SEM, STM/STS, Fourier transform spectroscopy (FTIR), Nuclear Magnetic Resonance (NMR) spectroscopy, Optically Detected Magnetic Resonance (ODMR) and Nitrogen-Vacancy (NV) center spectroscopy, Optical pumping and fluorescence measurement.
- Fluid Mechanics (Flow Measurement): Particle Image Velocimetry (PIV), Laser Doppler Anemometry (LDA).
- Materials Characterization: Single-crystal and powder X-ray diffraction.
- Analytical Chemistry : Solid-Phase Microextraction (SPME), Gas/Liquid Chromatography (GC, LC), Mass Spectrometry (MS).

ASSOCIATIVE RESPONSIBILITIES

Member of the ESPCI Student Council - responsible for the double degrees *March 2025 - March 2026*

- Accompanied 20 students including 12 international students (Brazil, Colombia, China) with their settlement, the administrative processes and the summer academic courses.

President of the ESPCI Welcome Club *March 2025 - March 2026*

- Organized induction events for double degree students.

INTERESTS

Soccer: 7 years in a club.

Volleyball: Volleyball Varsity Team Member, 4 years

Rock Climbing: 4 years in a club

Drawing: Courses at Nantes School of Fine Arts, (Sept 2019 - Jul 2021), Charcoal, Portraits, Painting, Photography.